

SMART WORKER HEALTH AND SAFETY MONITORING SYSTEM FOR CONSTRUCTION INDUSTRIES

A PROJECT REPORT

Submitted by

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1. Project Overview

The Smart Worker Health and Safety Monitoring System for Construction Industries is an IoT-based safety system designed to improve the health, safety, and protection of workers at construction sites. Construction industries involve several risks, including falls, excessive heat, hazardous gases, high physical stress, and sudden health emergencies. Continuous monitoring of workers can help identify dangerous conditions at an early stage and reduce the possibility of serious accidents.

The proposed system uses a combination of health, motion, and environmental sensors to continuously monitor important safety parameters. A wearable monitoring unit can be provided to each worker, which may include a heart-rate sensor, body-temperature sensor, gas sensor, accelerometer, and GPS module. These sensors collect real-time information about the worker and the surrounding environment. An ESP32/ESP8266 microcontroller processes the sensor data and compares the measured values with predefined safety limits.

When an abnormal condition is detected, the system immediately generates an alert. For example, if a worker falls or experiences a sudden impact, the accelerometer can identify the unusual movement and activate an emergency alert. Similarly, abnormal heart rate or body temperature can indicate physical stress or a possible health problem. The gas sensor can detect the presence of harmful gases, while environmental sensors can monitor conditions such as high temperature and humidity.

The system can provide local alerts through a buzzer or vibration motor so that the worker is immediately warned about the unsafe condition. At the same time, important information can be transmitted wirelessly to a site supervisor or monitoring station. The worker's location can also be identified using GPS during an emergency, allowing faster assistance and response.

The main purpose of this project is to provide real-time safety monitoring and early warning rather than depending only on manual supervision. By combining wearable technology, sensors, wireless communication, and automated alerts, the proposed system can improve emergency response and support better safety management at construction sites.

Overall, the Smart Worker Health and Safety Monitoring System provides a practical and cost-effective approach for protecting construction workers. The system can help reduce accident risks, improve workplace safety, and contribute to the development of a safer and smarter construction environment.

2. Problem Statement

Construction industries are among the most hazardous working environments because workers are continuously exposed to various physical, environmental, and operational risks. Construction sites involve activities such as working at heights, operating heavy machinery, handling construction materials, and working in areas with dust, heat, and hazardous gases. These conditions can lead to accidents, injuries, health problems, and, in severe cases, loss of life. Therefore, ensuring continuous monitoring of worker health and safety is an important requirement in modern construction industries.

One of the major problems is that conventional safety practices mainly depend on personal protective equipment and manual supervision. Although helmets, safety shoes, gloves, and safety harnesses provide protection, they cannot continuously monitor a worker's health condition or detect every dangerous situation. Supervisors may also be unable to observe every worker continuously, especially on large construction sites. As a result, emergencies such as sudden falls, excessive body temperature, abnormal heart rate, or exposure to harmful gases may not be detected immediately.

Another important problem is the delay in communicating emergency situations. When a worker experiences an accident or becomes unconscious, nearby workers or supervisors may not immediately know what has happened or where the incident occurred. This delay can increase the severity of injuries and reduce the effectiveness of emergency response.

Environmental conditions such as extreme temperature, humidity, and toxic gases can also affect worker health and performance. Prolonged exposure to such conditions may cause fatigue, dehydration, breathing difficulties, or other health-related problems. Without continuous monitoring, these risks may remain unnoticed until the worker's condition becomes serious.

Therefore, there is a need for an automated Worker Health and Safety Monitoring System that can continuously monitor worker health and environmental parameters, detect unsafe conditions, and generate immediate alerts. Such a system can provide early warnings to workers and supervisors, enable faster emergency response, reduce accident risks, and improve overall safety management in construction industries.

3. Proposed Solution

The proposed system is an IoT-based Worker Health and Safety Monitoring System designed to continuously monitor construction workers and their surrounding environment. A wearable device equipped with a heart-rate sensor, body-temperature sensor, gas sensor, accelerometer, and GPS module collects real-time information. An ESP32/ESP8266 microcontroller processes the sensor data and compares it with predefined safety limits. This allows the system to identify abnormal health conditions, hazardous gases, excessive temperature, and sudden falls.

When an unsafe condition is detected, the system immediately activates a buzzer or vibration alert to warn the worker. At the same time, the system can transmit the safety information wirelessly to a supervisor or monitoring station. GPS can help identify the worker's location during emergencies, enabling faster assistance. The proposed solution reduces dependence on manual supervision, improves emergency response, and provides a safer working environment for construction workers.

Main Features of the Proposed Solution

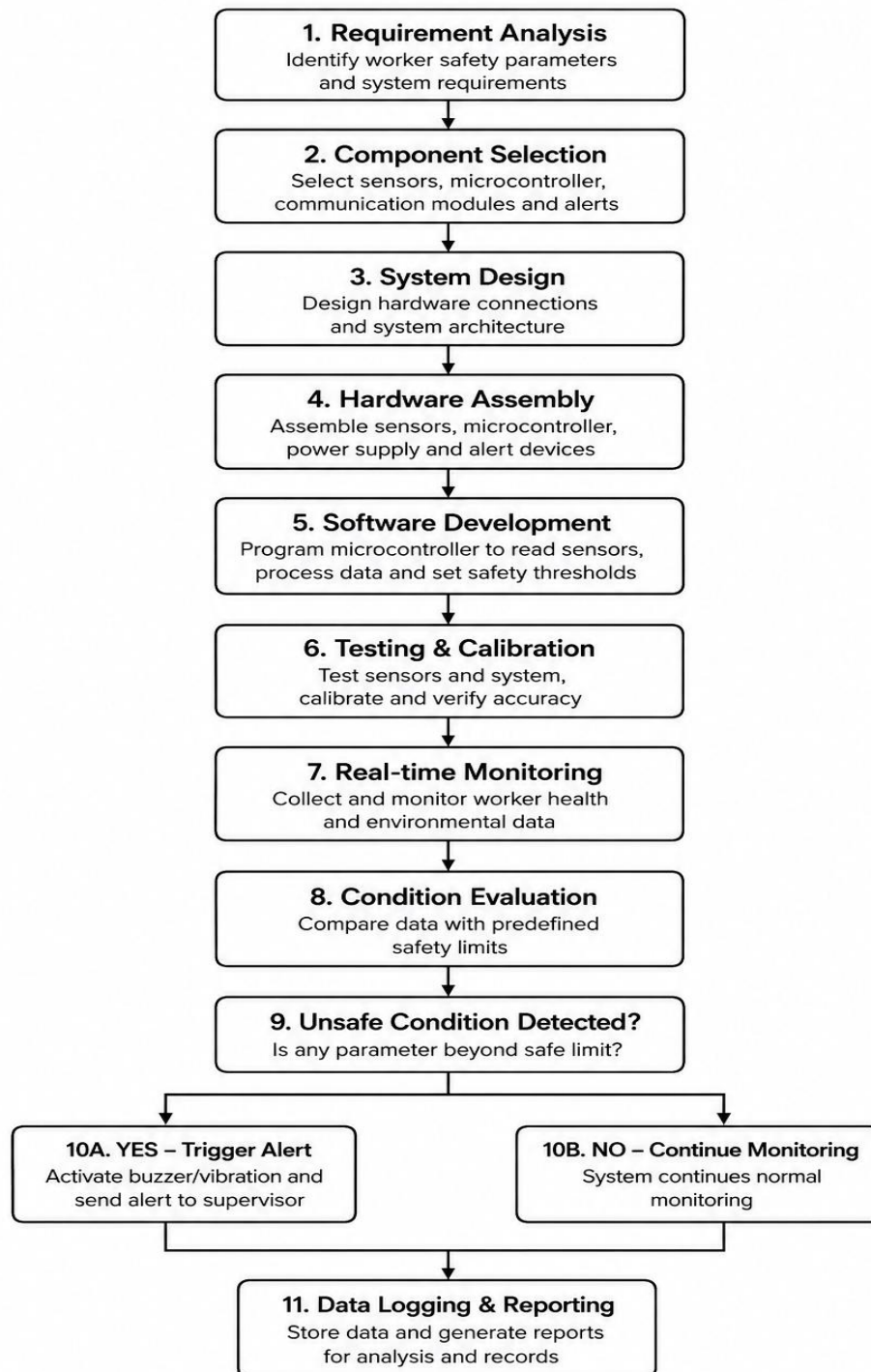
1. **Real-Time Health Monitoring** – Continuously monitors the worker's heart rate and body temperature.
2. **Fall Detection** – Uses an accelerometer to detect sudden falls, impacts, or abnormal movements.
3. **Hazardous Gas Detection** – Detects harmful or toxic gases present in the construction environment.
4. **Environmental Monitoring** – Monitors temperature and humidity to identify unsafe working conditions.
5. **Instant Alert System** – Activates a buzzer or vibration motor when dangerous conditions are detected.
6. **GPS Location Tracking** – Identifies the worker's location during emergencies for quick assistance.
7. **Wireless Communication** – Transmits real-time safety data to a supervisor or monitoring station.
8. **Automated Safety Monitoring** – Reduces dependence on continuous manual supervision and improves overall construction-site safety.

Overall Solution

The proposed system continuously monitors worker health and environmental conditions using sensors connected to an ESP32/ESP8266. It detects falls, abnormal heart rate, high temperature, and hazardous gases. When unsafe conditions occur, alerts are generated through a buzzer and wireless notification, enabling quick emergency response and improving construction-site safety.

4. System Architecture

The Smart Worker Health and Safety Monitoring System consists of sensors, a microcontroller, alert units, wireless communication, and a monitoring system.



5. Circuit Table

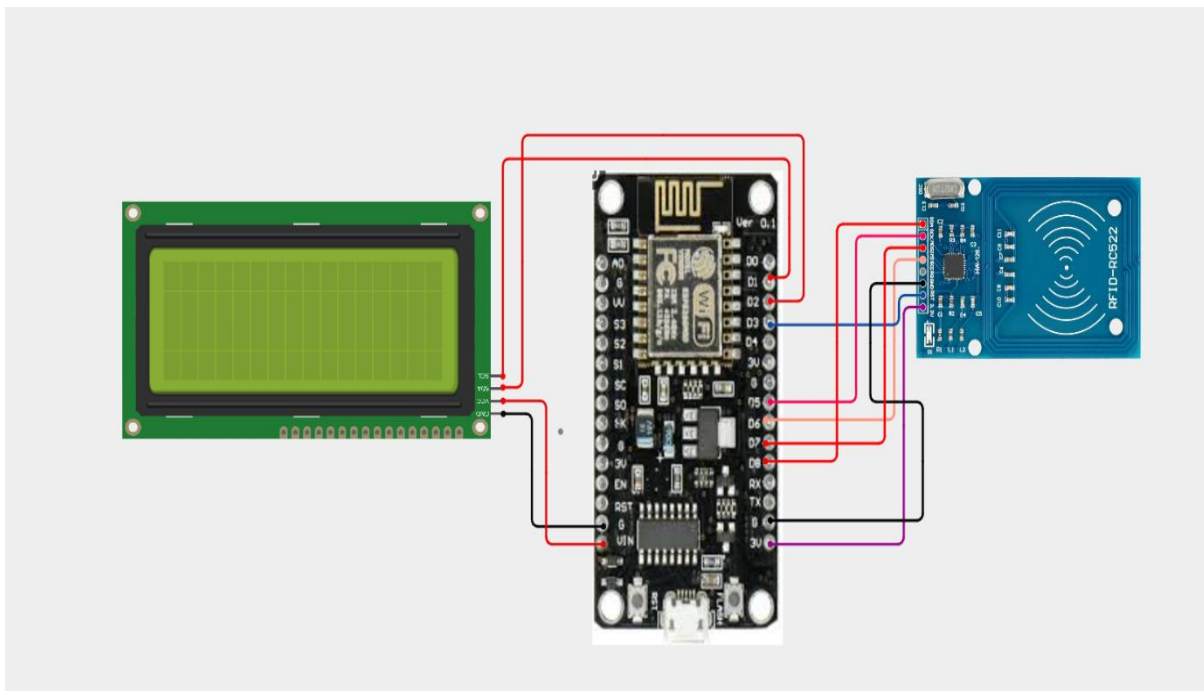
S.No.	Component	Pin/Terminal	ESP32 Pin	Function
1	Heart Rate Sensor	VCC	3.3V	Power supply
		GND	GND	Ground
		Signal	GPIO 34	Heart-rate measurement
2	Body Temperature Sensor (DS18B20)	VCC	3.3V	Power supply
		GND	GND	Ground
		DATA	GPIO 4	Body-temperature measurement
3	Gas Sensor (MQ-135)	VCC	5V/VIN	Power supply
		GND	GND	Ground
		AO	GPIO 35	Gas concentration monitoring
4	Accelerometer (MPU6050)	VCC	3.3V	Power supply
		GND	GND	Ground
		SDA	GPIO 21	I ² C data
		SCL	GPIO 22	I ² C clock
5	GPS Module (NEO-6M)	VCC	3.3V/5V*	Power supply
		GND	GND	Ground
		TX	GPIO 16 (RX2)	GPS data input
		RX	GPIO 17 (TX2)	GPS data output
6	Buzzer	+	GPIO 25	Unsafe-condition alert
		–	GND	Ground
7	Vibration Motor	+	GPIO 26*	Worker alert
		–	GND	Ground

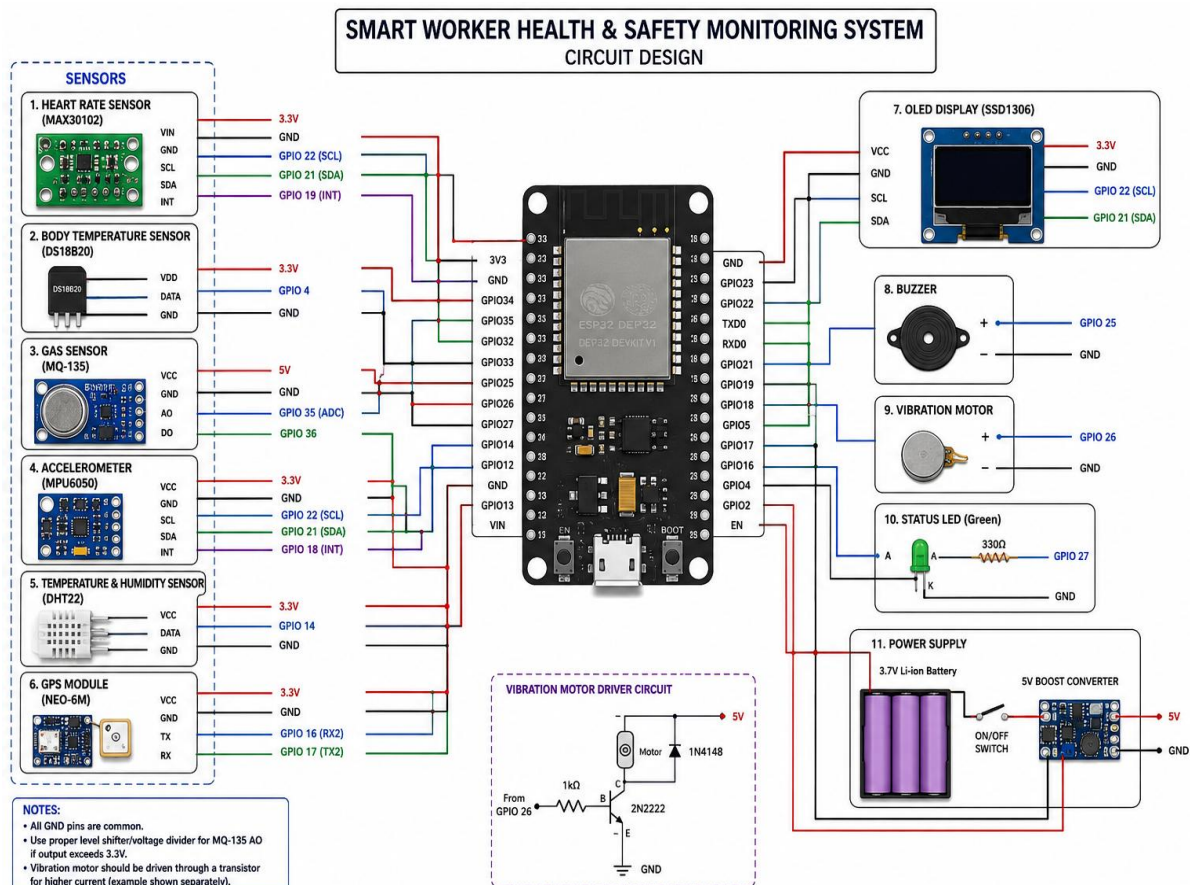
8	LED Indicator	Anode (+)	GPIO 27	Status indication
		Cathode (-)	GND through resistor	Ground
9	OLED Display	VCC	3.3V	Power supply
		GND	GND	Ground
		SDA	GPIO 21	I ² C data
		SCL	GPIO 22	I ² C clock
10	Battery/Power Supply	+	VIN/5V	System power
		-	GND	Common ground

Material Identification & Monitoring

- **ESP8266** → Processes the sensor and RFID data.
- **I2C LCD** → Displays material name and available quantity.
- **Buzzer** → Provides a low-stock or warning alert.
- **LED** → Indicates system/status conditions.

6. Circuit Design





7. Algorithm

- Start the system and initialize the ESP32 and all connected sensors.
- Collect data from the heart-rate, body-temperature, gas, accelerometer, and environmental sensors.
- Process the sensor data using the ESP32.
- Compare the readings with predefined safety limits.
- Check for unsafe conditions such as abnormal heart rate, high temperature, hazardous gas, or worker fall.
- If unsafe conditions are detected, activate the buzzer and vibration motor.
- Send an emergency alert and worker location to the supervisor through wireless communication.
- If conditions are normal, continue monitoring without triggering an alert.
- Display and store the sensor data for monitoring and future analysis.
- Repeat the monitoring process continuously.

8.Coding

```
#include <WiFi.h>

#include <HTTPClient.h>

#include <DHT.h>

#include <TinyGPS++.h>


// =====

// WIFI SETTINGS

// =====


const char* ssid = "No Hotspot";
const char* password = "sathish@2007";


// ThingSpeak Write API Key
String apiKey = "20QR1SF2BOGLN7I0";


// =====

// PIN DEFINITIONS

// =====


#define DHTPIN    4
#define DHTTYPE   DHT11


#define MQ2_PIN   27
#define BUZZER_PIN 26
#define PULSE_PIN 34


// GPS
#define GPS_RX    16 // ESP32 RX <- GPS TX
```

```
#define GPS_TX    17  // ESP32 TX -> GPS RX
```

```
// =====
```

```
// OBJECTS
```

```
// =====
```

```
DHT dht(DHTPIN, DHTTYPE);
```

```
TinyGPSPlus gps;
```

```
HardwareSerial gpsSerial(2);
```

```
// =====
```

```
// VARIABLES
```

```
// =====
```

```
float temperature = 0;
```

```
float humidity = 0;
```

```
int pulseValue = 0;
```

```
int bpm = 0;
```

```
bool beatDetected = false;
```

```
unsigned long lastBeat = 0;
```

```
bool gasDetected = false;
```

```
bool unsafe = false;
```

```

// =====
// THINGSPEAK TIMER
// =====

unsigned long lastThingSpeak = 0;

const unsigned long thingSpeakInterval = 20000;

// =====
// SETUP
// =====

void setup() {

  Serial.begin(115200);

  delay(1000);

  Serial.println();
  Serial.println("=====");
  Serial.println("  WORKER SAFETY MONITORING SYSTEM");
  Serial.println("=====");

  // =====
  // GPS
  // =====

  gpsSerial.begin(
    9600,

```

```

    SERIAL_8N1,
    GPS_RX,
    GPS_TX
);

Serial.println("GPS Serial Started");

// =====
// SENSOR PINS
// =====

pinMode(MQ2_PIN, INPUT);

pinMode(BUZZER_PIN, OUTPUT);

digitalWrite(BUZZER_PIN, LOW);

// =====
// DHT11
// =====

dht.begin();

Serial.println("DHT11 Started");

// =====
// WIFI
// =====

```

```

Serial.println();
Serial.print("Connecting to WiFi: ");
Serial.println(ssid);

WiFi.begin(
    ssid,
    password
);

while (
    WiFi.status() != WL_CONNECTED
) {

    delay(500);

    Serial.print(".");
}

Serial.println();
Serial.println("WiFi Connected!");

Serial.print("ESP32 IP Address: ");
Serial.println(WiFi.localIP());

Serial.println();
Serial.println("System Ready");
Serial.println("=====");
}

```

```

// =====
// LOOP
// =====

void loop() {

    // =====
    // DHT11
    // =====

    float newTemperature =
        dht.readTemperature();

    float newHumidity =
        dht.readHumidity();

    if (
        !isnan(newTemperature) &&
        !isnan(newHumidity)
    ) {

        temperature =
            newTemperature;

        humidity =
            newHumidity;
    }

    // =====

```

```

// MQ-2

// =====

gasDetected =
    (digitalRead(MQ2_PIN) == LOW);

// =====

// PULSE SENSOR

// =====

pulseValue =
    analogRead(PULSE_PIN);

unsigned long currentTime =
    millis();

// Simple pulse detection
//
// ESP32 ADC = approximately 0 to 4095
// Adjust threshold according to your sensor.

if (
    pulseValue > 2000 &&
    !beatDetected
) {

    beatDetected = true;

    if (lastBeat > 0) {

```

```

    unsigned long interval =
        currentTime - lastBeat;

    if (
        interval > 300 &&
        interval < 2000
    ) {

        bpm =
            60000 / interval;
    }
}

lastBeat =
    currentTime;
}

if (
    pulseValue < 1800
) {

    beatDetected = false;
}

// =====
// GPS
// =====

```



```
while (
    gpsSerial.available()
) {

    gps.encode(
        gpsSerial.read()
    );
}

// =====
// SAFETY CHECK
// =====

unsafe = false;

// Temperature
if (
    temperature > 40
) {

    unsafe = true;
}

// Humidity
if (
    humidity > 80
) {

    unsafe = true;
```

```

    }

    // Gas
    if (
        gasDetected
    ) {

        unsafe = true;
    }

    // Heart rate
    if (
        bpm > 120
    ) {

        unsafe = true;
    }

    // =====
    // BUZZER
    // =====

    if (unsafe) {

        digitalWrite(
            BUZZER_PIN,
            HIGH
        );
    }

```

```

    } else {

        digitalWrite(
            BUZZER_PIN,
            LOW
        );
    }

    // =====

    // SERIAL MONITOR

    // =====

    Serial.println();
    Serial.println("-----");

    Serial.print("Temperature : ");
    Serial.print(temperature, 1);
    Serial.println(" °C");

    Serial.print("Humidity  : ");
    Serial.print(humidity, 1);
    Serial.println(" %");

    Serial.print("Heart Rate : ");
    Serial.print(bpm);
    Serial.println(" BPM");

    Serial.print("Pulse ADC  : ");
    Serial.println(pulseValue);

```

```

Serial.print("Gas Status : ");

if (gasDetected) {

    Serial.println("DANGER");

} else {

    Serial.println("SAFE");
}

// =====
// GPS OUTPUT
// =====

if (
    gps.location.isValid()
) {

    Serial.print("GPS Status : FIXED");

    Serial.println();

    Serial.print("Latitude   : ");
    Serial.println(
        gps.location.lat(),
        6
    );

```

```

Serial.print("Longitude  : ");
Serial.println(
    gps.location.lng(),
    6
);

} else {

    Serial.println(
        "GPS Status : SEARCHING..."
    );
}

// =====
// SATELLITES
// =====

if (
    gps.satellites.isValid()
) {

    Serial.print("Satellites  : ");
    Serial.println(
        gps.satellites.value()
    );
}

// =====

```

```

// SAFETY STATUS

// =====

Serial.print("Safety   : ");

if (unsafe) {

    Serial.println("!!! UNSAFE !!!");

} else {

    Serial.println("SAFE");
}

// =====

// WIFI STATUS

// =====

Serial.print("WiFi     : ");

if (
    WiFi.status() == WL_CONNECTED
) {

    Serial.println("CONNECTED");

} else {

    Serial.println("DISCONNECTED");
}

```

```

}

Serial.println("-----");

// =====
// THINGSPEAK UPDATE
// =====

if (
  millis() - lastThingSpeak >=
  thingSpeakInterval
) {

  lastThingSpeak =
    millis();

  sendToThingSpeak();
}

delay(1000);
}

// =====
// THINGSPEAK FUNCTION
// =====

void sendToThingSpeak() {

  Serial.println();

```

```
Serial.println(">>> Sending data to ThingSpeak...");
```

```
if (
```

```
    WiFi.status() != WL_CONNECTED
```

```
) {
```

```
    Serial.println(
```

```
        "ERROR: WiFi disconnected"
```

```
    );
```

```
    return;
```

```
}
```

```
HTTPClient http;
```

```
String url =
```

```
    "http://api.thingspeak.com/update?api_key=";
```

```
url += apiKey;
```

```
// =====
```

```
// FIELD 1 - TEMPERATURE
```

```
// =====
```

```
url += "&field1=";
```

```
url += String(
```

```
    temperature,
```

```
    1
```

```
);
```



```
// =====  
// FIELD 2 - HUMIDITY  
// =====
```

```
url += "&field2=";  
url += String(  
    humidity,  
    1  
);
```

```
// =====  
// FIELD 3 - HEART RATE  
// =====
```

```
url += "&field3=";  
url += String(  
    bpm  
);
```

```
// =====  
// FIELD 4 - GAS  
// =====
```

```
url += "&field4=";
```

```
if (gasDetected) {
```

```
    url += "1";
```

```

} else {

    url += "0";
}

// =====
// FIELD 5 - LATITUDE
// =====

if (
    gps.location.isValid()
) {

    url += "&field5=";

    url += String(
        gps.location.lat(),
        6
    );

    // =====
    // FIELD 6 - LONGITUDE
    // =====

    url += "&field6=";

    url += String(
        gps.location.lng(),

```

```
        6
    );
}

// =====
// FIELD 7 - SAFETY
// =====

url += "&field7=";

if (unsafe) {

    url += "1";

} else {

    url += "0";
}

// =====
// SEND DATA
// =====

http.begin(url);

int httpCode =
    http.GET();

if (httpCode > 0) {
```

```
Serial.print(  
    "ThingSpeak HTTP Code: "  
);
```

```
Serial.println(  
    httpCode  
);
```

```
String response =  
    http.getString();
```

```
Serial.print(  
    "ThingSpeak Entry ID: "  
);
```

```
Serial.println(  
    response  
);
```

```
} else {
```

```
Serial.print(  
    "ThingSpeak ERROR: "  
);
```

```
Serial.println(  
    httpCode  
);
```

```

}

http.end();

Serial.println(
  ">>> ThingSpeak update complete"
);
}

```

9. System Working

The Smart Worker Health and Safety Monitoring System is designed to continuously monitor the health of construction workers and the environmental conditions around them. The system uses an ESP32 microcontroller as the main processing unit and several sensors connected to a wearable safety device. The sensors collect real-time information and send it to the ESP32, where the data is processed and compared with predefined safety limits.

1. Sensor Data Collection

When the system is switched ON, the ESP32 initializes all the connected sensors. The MAX30102 heart-rate sensor continuously measures the worker's heart rate, while the DS18B20 temperature sensor monitors body temperature. The DHT22 sensor measures the surrounding temperature and humidity. These parameters help identify physical stress and uncomfortable or unsafe environmental conditions. The MQ-135 gas sensor monitors the surrounding air for harmful gases. This is particularly useful in construction areas where workers may be exposed to polluted air, chemical fumes, or other hazardous gases. The MPU6050 accelerometer continuously monitors the worker's movement and detects sudden changes in acceleration that may indicate a fall or strong impact.

2. Data Processing

The sensor readings are continuously transferred to the ESP32. The microcontroller processes the collected data and compares each parameter with predefined safety limits. For example, if the heart rate becomes unusually high or low, the system identifies it as a possible health warning. Similarly, excessive body temperature or extreme environmental temperature can indicate heat stress. The accelerometer data is analyzed to identify sudden falls or abnormal movements. If a significant impact followed by a lack of normal movement is detected, the system can classify the event as a possible fall. The gas sensor reading is also compared with a predefined threshold to determine whether the surrounding air has reached a dangerous level.

3. Safety Condition Evaluation

After processing the sensor readings, the ESP32 evaluates whether the worker is operating under safe or unsafe conditions. If all measured parameters remain within their defined limits, the system continues normal monitoring. The current values can be displayed on the OLED display, allowing the worker or supervisor to observe important safety information. If one or more parameters exceed the safe limit, the system identifies the situation as an unsafe condition. Different types of unsafe conditions can include abnormal heart rate, excessive body temperature, hazardous gas concentration, extreme environmental conditions, or sudden worker falls.

4. Local Alert Generation

When an unsafe condition is detected, the system immediately activates the buzzer and vibration motor. The buzzer provides an audible warning, while the vibration motor provides a direct alert to the worker. An LED indicator can also be activated to show that a warning condition has occurred. For example, if a hazardous gas concentration is detected, the worker receives an immediate warning so that appropriate safety action can be taken. In the case of a detected fall, the alert can indicate that the worker may require immediate assistance.

5. Location Tracking and Emergency Communication

The NEO-6M GPS module provides the geographical location of the worker. During an emergency, the worker's location can be obtained and included in the safety information sent to the monitoring system. This is especially useful on large construction sites where locating an injured or unconscious worker may otherwise take considerable time. The ESP32 can use Wi-Fi or another suitable wireless communication method to transmit sensor information and emergency status to a monitoring platform. The supervisor can then receive information about the worker's condition and take appropriate action.

6. Supervisor Monitoring

The collected information can be displayed on a monitoring dashboard or mobile/web-based interface. The supervisor can monitor parameters such as heart rate, body temperature, environmental temperature, humidity, gas level, and worker location. The monitoring system can distinguish between normal and warning conditions. In an emergency, the supervisor receives an alert and can identify the type of problem and the worker involved. This helps reduce the response time and improves emergency management at the construction site.

7. Continuous Monitoring

The system does not stop after checking a single sensor reading. It continuously collects and processes new data at regular intervals. This creates a real-time safety monitoring system in which changes in the worker's health or surrounding environment can be detected quickly.

If the condition returns to the normal range, the system can deactivate the warning and return to normal monitoring. If the unsafe condition continues, the warning remains active until appropriate action is taken.

8. Data Recording and Analysis

Important sensor readings and safety events can be recorded for future analysis. The stored information can help supervisors identify repeated safety problems, environmental hazards, or periods of excessive worker stress. The recorded data can also support preventive safety measures. For example, if high temperatures are repeatedly detected during a particular working period, supervisors can introduce additional breaks, hydration arrangements, or changes to the work schedule.

10. Bill of Materials

Since this is a software system (not a physical product), the BOM here covers the hardware, software, and services needed to build and deploy it — rather than construction materials (which the system itself tracks via the BOQ).

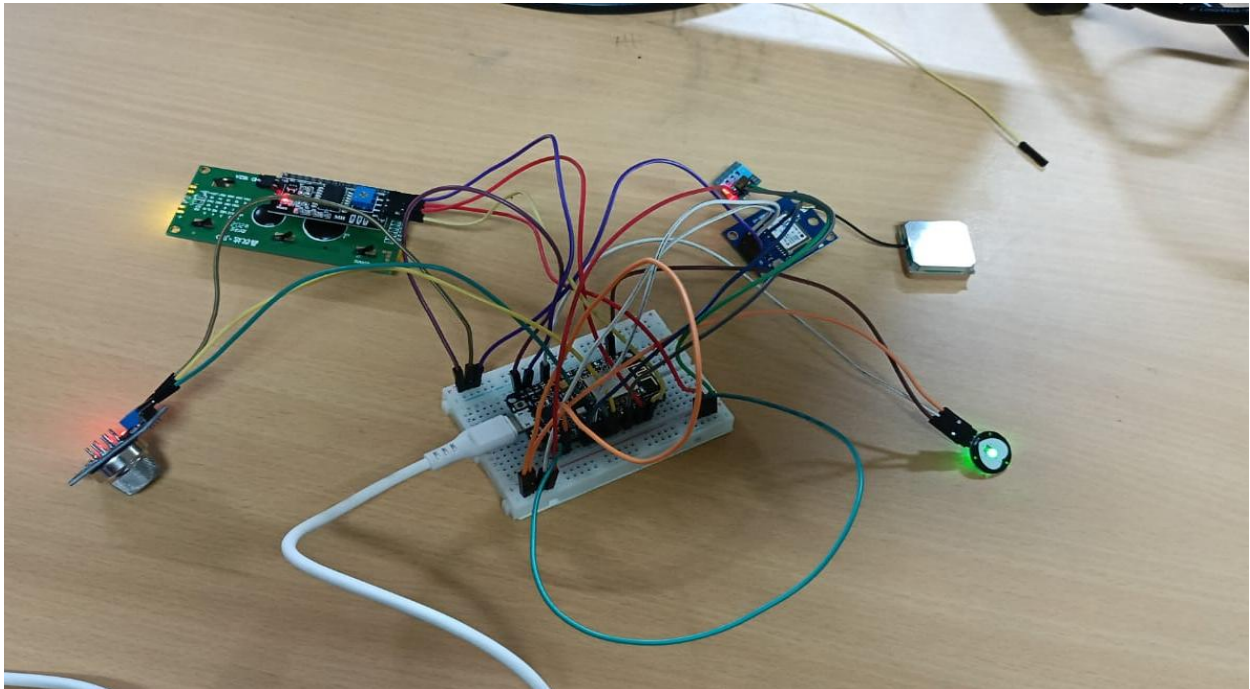
S.No.	Component	Specification	Quantity
1	ESP32 Development Board	ESP32-WROOM-32	1
2	Heart Rate Sensor	MAX30102	1
3	Body Temperature Sensor	DS18B20	1
4	Gas Sensor	MQ-135	1
5	Accelerometer	MPU6050	1
6	Temperature & Humidity Sensor	DHT22	1
7	GPS Module	NEO-6M	1
8	OLED Display	0.96-inch I ² C, SSD1306	1
9	Buzzer	5 V Active Buzzer	1
10	Vibration Motor	3 V DC	1
11	LED	5 mm LED	2
12	Resistors	220 Ω , 4.7 k Ω	As required
13	Transistor/MOSFET	Motor driver	1
14	Diode	1N4007	1
15	Battery	Rechargeable Li-ion	1

S.No.	Component	Specification	Quantity
16	Boost Converter	5 V Output	1
17	Breadboard/PCB	Prototype board	1
18	Jumper Wires	Male–Male/Female–Male	As required
19	Switch	ON/OFF	1
20	Safety Helmet/Enclosure	Prototype mounting	1

11. Prototype Implementation

- **ESP32 Setup** – The ESP32 is used as the main controller for processing all sensor data.
- **Sensor Integration** – Connect the MAX30102, DS18B20, MQ-135, MPU6050, DHT22, and GPS sensors to the ESP32.
- **Health Monitoring** – The MAX30102 measures heart rate and the DS18B20 monitors body temperature.
- **Environmental Monitoring** – The MQ-135 detects harmful gases, while the DHT22 measures temperature and humidity.
- **Fall Detection** – The MPU6050 detects sudden movements, impacts, and possible worker falls.
- **Location Tracking** – The NEO-6M GPS module provides the worker's location during emergencies.
- **Display Unit** – The OLED display shows sensor readings and the current safety status.
- **Alert System** – A buzzer and vibration motor warn the worker when an unsafe condition is detected.
- **Wireless Communication** – ESP32 transmits safety information to a supervisor or monitoring system.
- **Power Supply** – A rechargeable battery provides power to the wearable prototype.
- **Programming** – The ESP32 is programmed to continuously read sensor values and compare them with predefined safety limits.
- **Testing** – The prototype is tested under simulated conditions such as high temperature, gas detection, abnormal movement, and fall detection.

- Final Prototype – All components are assembled into a compact wearable unit suitable for mounting on a safety helmet, vest, or belt.



12.Result and Performance Analysis

The developed Smart Worker Health and Safety Monitoring System successfully monitors important worker health and environmental parameters in real time. The ESP32 collects data from the heart-rate, body-temperature, gas, accelerometer, temperature/humidity, and GPS sensors. When an unsafe condition such as a fall, abnormal health reading, excessive temperature, or hazardous gas level is detected, the system activates the buzzer and vibration motor to alert the worker. The prototype also displays the monitored parameters through the OLED display and can transmit safety information wirelessly to the supervisor. The GPS module provides the worker's location during emergency situations. Overall, the prototype demonstrates the feasibility of using IoT technology for continuous construction-worker safety monitoring.

Performance Analysis

Parameter	Expected Performance	Observation
Heart-rate monitoring	Continuous measurement	Health status can be monitored in real time
Body-temperature monitoring	Continuous measurement	Abnormal temperature can generate an alert

Parameter	Expected Performance	Observation
Gas detection	Hazard detection	Warning generated when gas level exceeds threshold
Fall detection	Sudden impact/movement detection	Alert generated during simulated falls
Temperature & humidity	Environmental monitoring	Helps identify uncomfortable/unsafe conditions
GPS tracking	Location identification	Worker location available during emergencies
Alert system	Fast warning	Buzzer and vibration provide immediate local alerts
OLED display	Real-time indication	Sensor values and status can be displayed
Wireless communication	Data transmission	Safety information can be sent to monitoring system
Overall operation	Continuous monitoring	System operates as an integrated safety unit

Overall Performance

The prototype provides real-time monitoring, automatic hazard detection, and immediate alert generation. The integration of multiple sensors with the ESP32 makes the system suitable for construction-site safety applications. Performance can be further improved by calibrating the sensors, optimizing threshold values, improving wireless connectivity, and using a compact PCB-based wearable design.

